

RESEARCH ARTICLE

1-Ethyl-3-Methylimidazolium Tetrafluoroborate Ionic Liquid as an Efficient Catalyst for the Methanolysis of PET

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ABSTRACT

The recycling of polyethylene terephthalate (PET), a widely used polyester plastic, has become an increasingly significant issue. This study aims to develop an efficient and environmentally friendly catalytic system for the degradation of waste PET and realize the resource utilization of polyester. The methanolysis of PET was investigated using 1-ethyl-3-methylimidazolium tetrafluoroborate ([Emim]BF₄) as a catalyst. The effects of temperature, time, [Emim]BF₄ dosage, and methanol dosage on the reaction were systematically examined. The conversion of PET reached 100% with a dimethyl terephthalate (DMT) yield of 74.06% under optimal conditions: 200°C, 2 h, a PET-to-[Emim]BF₄ mass ratio of 5:1, and a PET-to-methanol mass ratio of 6:1. Kinetic analysis revealed that the reaction followed first-order kinetics, with an apparent activation energy of 51.16 kJ/mol. A possible mechanism for the [Emim]BF₄-catalyzed methanolysis of PET was proposed. Furthermore, density functional theory (DFT) calculations demonstrated that [Emim]BF₄ could effectively reduce both the reaction energy barriers and the final-state energy, thereby enhancing the reaction rate and favoring the forward progression of the reaction.

1 | Introduction

Polyethylene terephthalate (PET) is a semi-crystalline thermoplastic polyester with excellent mechanical properties, stable chemical performance, and high transparency. It has become an indispensable material in the plastics packaging, textile, construction, healthcare, and other fields [1, 2]. According to statistics, the global annual production of PET exceeded 26 million tons in 2024, showing a continuously significant upward trend [3]. However, PET is difficult to be naturally degraded, and its recycling rate is generally below 10% [4, 5]. A large amount of discarded PET continues to accumulate in the environment, causing severe environmental pollution and resource waste problems. Therefore, developing efficient technologies for the recycling and reuse of waste PET is of great significance for reducing environmental burdens and promoting resource circulation.

Chemical recycling is one of the mainstream methods for recycling PET. Its core lies in depolymerizing PET into monomers through chemical means; the products can be used to synthesize high value-added chemicals after purification, representing a true circular economy pathway [6]. The main chemical recycling methods include glycolysis, hydrolysis, methanolysis, and ammonolysis. Methanolysis can depolymerize PET into dimethyl terephthalate (DMT) and ethylene glycol (EG) through a relatively simple process without colored byproducts produced [7]. The monomer DMT can be easily purified and used to produce compounds with high economic value, such as Polyethylene Terephthalate Glycol (PETG) and 1,4-Cyclohexanedimethanol (CHDM) [8]. However, uncatalyzed methanolysis has low yield and long reaction time, while supercritical methanolysis has the drawback of harsh reaction conditions. In this context, catalytic methanolysis, which offers high yield, fast reaction rate, and relatively mild reaction conditions, has gradually attracted attention.

Common catalysts for PET alcoholysis include metal salts, metal oxides, ionic liquids, and deep eutectic solvents [9, 10]. Although many of these catalysts exhibit excellent catalytic performance, they often have problems such as high cost, strong toxicity, high corrosiveness, and potential environmental pollution. Therefore, developing catalysts that are efficient, environmentally friendly, and economically feasible has become a key challenge in advancing PET alcoholysis technology.

Ionic liquids, composed of cations and anions or ionic groups, are a class of organic salts that remain liquid at or near room temperature with extremely low volatility. They facilitate the alcoholysis of PET under mild conditions within a short time frame, owing to their excellent thermal stability, structural tunability, and unique solvation properties, while also allowing for easy separation from the solid products [11, 12]. In 2009, Wang et al. [13] used a series of 1-butyl-3-methylimidazolium halide ionic liquids (e.g., [Bmim]Cl and [Bmim]Br) as catalysts to depolymerize PET to bis(2-hydroxyethyl) terephthalate (BHET), marking the initial application of ionic liquids in PET glycolysis. To address the limitations such as long reaction time, low product selectivity, and poor yield, they subsequently synthesized an iron-containing ionic liquid [Bmim]FeCl₄ and attained 100% of PET conversion and 59.2% of BHET selectivity at 178°C and 4 h [14]. The enhanced catalytic activity of [Bmim]FeCl₄ compared to FeCl₃ and [Bmim]Cl alone was attributed to the synergistic effect between the cations and anions.

The catalytic performance of ionic liquids is highly dependent on the specific structures of their constituent cations and anions. For instance, Yue et al. [15] synthesized the Lewis acidic ionic liquid [Bmim]ZnCl₃ for PET glycolysis. Under optimal conditions (0.31 wt% [Bmim]ZnCl₃, 2 g PET, 22 g EG, 190°C, 2 h, atmospheric pressure), a PET conversion of 100% and a 79.5% yield of BHET were achieved. Wang et al. [16] reported that 1,3-diethylimidazolium triacetate zincate ([deim][Zn(OAc)₂]) could act as an efficient catalyst in PET glycolysis [16]. Furthermore, Liu et al. [17, 18] developed a series of Brønsted–Lewis acidic ionic liquids, [HO₃S-(CH₂)₃-NEt₃][Cl-[ZnCl₂]_x] and [HO₃S-(CH₂)₃-NEt₃][Cl-[FeCl₃]_x] (where *x* represents the molar fraction of the metal chloride), and applied them to catalyze the degradation of PET in *n*-butanol. The results indicated that these ionic liquids possess a greater number of acid sites, with their high activity attributed to a synergistic effect between Brønsted and Lewis acids. Nevertheless, the industrial application of such ionic liquids faces challenges, including potential environmental pollution from metal ions, negative effects on product quality, complex synthesis procedures, and high costs.

In parallel, the catalytic potential of metal-free ionic liquids has also been explored. Al-Sabagh et al. [19] investigated the glycolysis of PET catalyzed by the Lewis basic ionic liquid [Bmim][OAc], and achieved a BHET yield of 58.2% under optimal conditions of 1 g [Bmim][OAc], 3 g PET, 20 g EG, 190°C, and 3 h. Liu et al. [20] reported that choline acetate ([Ch][OAc]) exhibited remarkable catalytic activity and reusability in PET glycolysis, with a high BHET yield of 85.2%. Nunes et al. [21] investigated the catalytic role of [Bmim]BF₄ in the supercritical ethanolysis of PET and suggested that the [Bmim]⁺/[BF₄]⁻ pair, in addition to supercritical ethanol, holds considerable promise for catalytic

PET depolymerization. In contrast, studies on ionic liquid-catalyzed methanolysis of PET remain scarce.

In this study, the catalytic performance of metal-free ionic liquid [Emim]BF₄ in the methanolysis of PET was examined. The stability of [Emim]BF₄ and the structure and purity of the degradation products were characterized by high performance liquid chromatography (HPLC), Fourier transform infrared (FT-IR) spectrometry, nuclear magnetic resonance (NMR) spectroscopy, differential scanning calorimetry (DSC), thermogravimetric analysis (TGA), and scanning electron microscope (SEM). After the reaction conditions were optimized, the high conversion of PET and yield of DMT were obtained. Moreover, a mechanism for the reaction is proposed, and the effect of [Emim]BF₄ on the methanolysis of PET was studied through DFT calculations.

2 | Experimental Section

2.1 | Materials

PET bottle flakes (average size 3 mm × 3 mm) were obtained from waste Nongfu Spring mineral water bottles, washed with deionized water, and dried in an oven at 60°C for 24 h. Anhydrous methanol (≥98% purity) was purchased from Tianjin Kemio Chemical Reagent Co. Ltd. 1-ethyl-3-methylimidazolium tetrafluoroborate ([Emim]BF₄) (99% purity) was purchased from Shanghai Chengjie Chemical Co. Ltd. (China). DMT (≥99% purity), 2-hydroxyethyl methyl terephthalate (MHET) (>97%), and BHET (>85% purity) were purchased from Aladdin reagent (Shanghai) Ltd.

2.2 | Methanolysis of PET Catalyzed by [Emim]BF₄

Predetermined amounts of PET, methanol, and [Emim]BF₄ were charged into a 50 mL reaction kettle with a maximum working pressure of 25 MPa (Figure S1), which was subsequently sealed. Prior to the reaction, the air inside the kettle was purged with nitrogen. The methanolysis was carried out with continuous magnetic stirring in an oil bath, with the temperature maintained at 170°C–220°C for 0.5–3.0 h. Upon completion, the reaction mixture was quickly cooled to ambient temperature. Solid–liquid separation was achieved by vacuum filtration. The filter cake was dried to constant weight at 60°C in an oven, after which it was weighed and collected as the degradation product for further analysis. EG, [Emim]BF₄, and residual methanol in the filtrate were separated by rotary evaporation for recycling and reusing. Then the products and the catalyst were characterized by a suite of analytical techniques, including HPLC, FT-IR spectrometry, NMR spectroscopy, DSC, TGA, and SEM.

A certain amount of the products was dissolved in excessive methanol and subsequently filtered to separate the undepolymerized PET. The residual PET was washed with distilled water, dried, and then weighed. The conversion of PET is expressed by Equation (1):

$$\text{Conversion} = \frac{m_{i,\text{PET}} - m_{f,\text{PET}}}{m_{i,\text{PET}}} \times 100\% \quad (1)$$

where $m_{i,PET}$ is the initial weight of PET and $m_{f,PET}$ is the final weight of the undepolymerized PET.

The quantitative analysis of monomers such as DMT, BHET, and MHET was performed using HPLC with the external standard method. For analysis, 0.004 g of the depolymerization products was dissolved in 100 mL of a methanol/deionized water mixture (7:3, v/v). The yields of monomers are defined as Equation (2):

$$\text{Yield} = \frac{m_{f,\text{mon}} / M_{\text{mon}}}{m_{f,PET} / M_{PET}} \times 100\% \quad (2)$$

where $m_{f,\text{mon}}$ are the final weights of monomers, which are calculated according to the corresponding peak areas obtained from the HPLC chromatogram. M_{PET} and M_{mon} are the molecular weights of the PET repeat unit (192 g/mol) and monomers such as DMT (194 g/mol), BHET (254 g/mol), and MHET (224 g/mol).

2.3 | Characterization and Instruments

TGA of the degradation products and [Emim]BF₄ were performed at a heating rate of 10°C/min in a nitrogen atmosphere using a Q600 thermogravimetric (TG) analyzer (TPA Instruments, USA) during a temperature range from ambient temperature to 600°C. DSC scan of the products and [Emim]BF₄ was obtained using DSC300 (NETZSCH, Germany) at a rate of 10°C/min in a nitrogen atmosphere during a temperature range from room temperature to 300°C and 600°C, respectively. The composition and main functional groups of the products were determined with an iS50 FT-IR spectrometer (Thermo Fisher Scientific, USA) in the range of 7800 to 100 cm⁻¹. The full-spectrum linear accuracy was better than 0.07% T, and the resolution was better than 0.09 cm⁻¹. ¹H NMR spectra of the products were recorded by nuclear magnetic resonance using an AVANCE NEO 600M NMR spectrometer (Bruker, USA) at 600 MHz in deuterated methanol (CD₃OD) solution and calibrated with tetramethylsilant (TMS) as the internal standard. HPLC analysis of the products was performed using an Agilent-1100 HPLC (Agilent, USA) equipped with a C8 column

(4.6 mm diameter and 150 mm long) and a SPD-1 ultraviolet detector set at 254 nm. The chromatographic conditions were as follows: mobile phase of 70% methanol and 30% ultrapure water, flow rate of 5 μL, and column temperature of 25°C. Microstructure morphology of PET at different stages of the reaction was examined by a FlexSEM1000 Microscope (Hitachi High-Tech, Japan) at an accelerating voltage of 10.0 kV, working distance of 7.7 mm, and magnification of 1.50 k, using secondary electron (SE) imaging.

2.4 | DFT Calculations

The simulation calculations were carried out based on the Materials Studio (MS) software platform, employing the DMol3 module with the generalized gradient approximation (GGA) functional and double numerical polarization (DNP) basis set in combination with the Perdew–Burke–Ernzerhof (PBE) functional. Through geometry optimization and transition state search, and by comparing the energy barrier changes in systems with and without ionic liquid, the mechanism of PET methanolysis catalyzed by [Emim]BF₄ was systematically investigated.

3 | Results and Discussion

3.1 | Characterization of the Catalyst and Products

The thermal stability of [Emim]BF₄, a prerequisite for its catalytic application, was first evaluated by TGA and DSC over a temperature range of 25°C–600°C (Figure 1). The weight and derivative weight curves were shown in Figure 1a, and the DSC curve was shown in Figure 1b. It could be observed that [Emim]BF₄ was very stable when the temperature was lower than 280°C. The weight began to slightly decrease when the temperature exceeded 280°C. Above 350°C, the weight curve showed a sharp downward trend, indicating the main decomposition stage, during which the molecular structure of [Emim]BF₄ broke down and gaseous products were released. The similarity in peak temperatures between the derivative weight and DSC curves suggested that both techniques

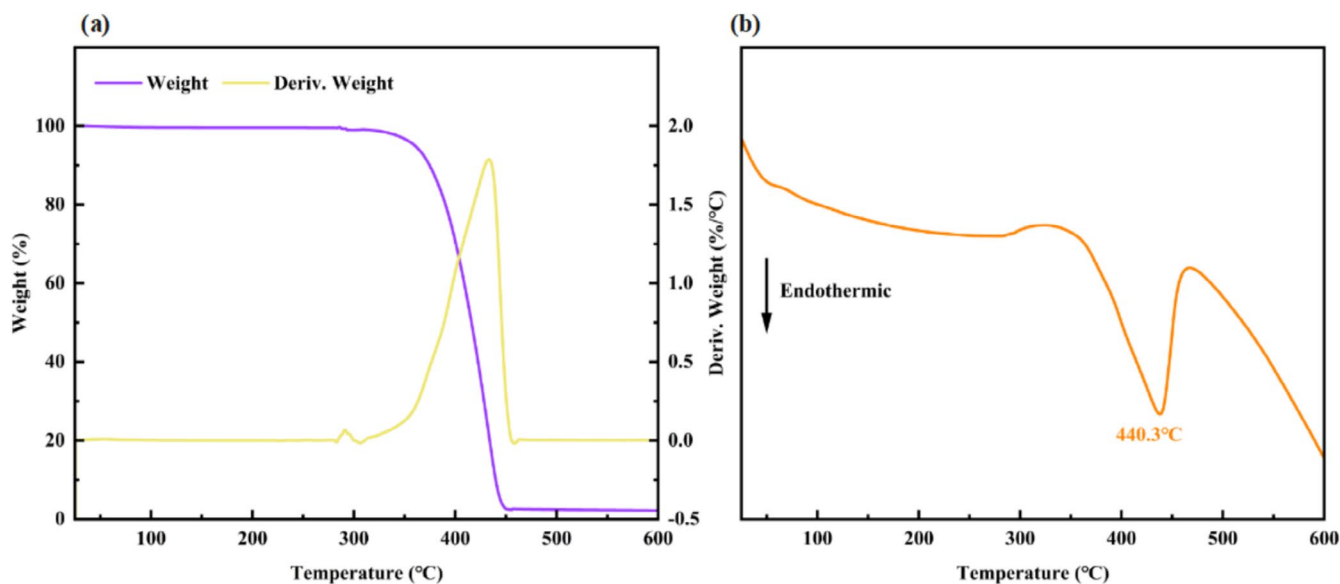


FIGURE 1 | (a) TGA curves and (b) DSC curve of [Emim]BF₄. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.70802)]

reflected the same intensive thermal decomposition process. Beyond 460°C, the mass stabilizes, indicating the completion of decomposition. The combined TGA-DSC results demonstrated that [Emim]BF₄ exhibited sufficient thermal stability to meet the temperature requirements for catalytic PET methanolysis.

Having confirmed the stability of the catalyst, the chemical composition of the solid depolymerization products was then characterized by FT-IR and ¹H NMR while the purity was determined by HPLC. Figure 2 showed similar FT-IR spectra of the degradation products and standard DMT, indicating the purity. The absorption peaks observed at 2959.3 cm⁻¹ were attributed to C–H stretching vibrations of groups, while the strong bands appearing at 1715.7 cm⁻¹ corresponded to the carbonyl (C=O) stretching vibrations [22]. The peaks observed at 1506.5 cm⁻¹ arose from the aromatic skeleton vibrations [19, 23]. Additionally, the absorption bands at 1386.1 cm⁻¹ were assigned to the bending vibrations of the methyl groups [11]. It was noteworthy that the positions and intensities of these characteristic peaks showed no significant differences from those of the reference DMT spectrum, especially at 200°C. These results indicated that PET was nearly completely depolymerized under all three reaction temperatures, and the main product DMT was obtained.

The chemical structure was further confirmed by ¹H NMR spectroscopy, as shown in Figure 3. The signal at δ = 8.10 ppm indicated the presence of four aromatic protons of the benzene ring [24]. The main peak observed at δ = 3.94 ppm corresponds to the methyl protons (–CH₃) adjacent to the ester groups (–COO–) [25]. The spectrum was in excellent agreement with that of the standard DMT (Figure S2), with no noticeable impurity peaks. Furthermore, the integral ratio of the two main peaks (a and b) is 1:1.51, closely matching the theoretical ratio of DMT. These results collectively confirm that the main component of the products is DMT with high purity.

Qualitative analysis by HPLC revealed that the main product was DMT (Figure 4). The HPLC results of the standard DMT,

MHET, and BHET are shown in Figures S3–S5. The products exhibited a retention time (3.245 min) identical to that of the standard DMT under the same chromatographic conditions. Although minor peaks corresponding to oligomers and the intermediate MHET were detected, the DMT peak was dominant in both height and area. The yield of individual components was determined based on their corresponding peak areas.

Finally, the thermal properties of the product were examined by DSC (Figure 5). Both samples exhibited sharp endothermic peaks, at 143.1°C for the products and 146.3°C for the standard DMT, corresponding to their respective melting transitions. The close proximity of these melting points indicated similar crystallinity and crystal structure between them. Furthermore, broader endothermic events were observed at higher temperatures (220.3°C for the product and 253.6°C for standard DMT), which were plausibly attributed to thermal transitions such as polymorphic transformation, thermal decomposition, or sublimation of the samples. The depression in the two peak temperatures of the products might be attributable to the presence of trace amounts of oligomers and intermediates.

3.2 | Parameter Optimization for Efficient PET Methanolysis

To optimize the reaction conditions, the effect of temperature, time, catalyst dosage, and methanol dosage was investigated. To maintain consistency, certain experimental parameters were kept constant, including a reaction temperature of 200°C, a reaction time of 2 h, a PET-to-IL mass ratio of 5:1, and a PET-to-methanol mass ratio of 1:6.

The effect of temperature on the conversion of PET and the yield of DMT was investigated, with the results presented in Figure 6a. It was observed that no significant depolymerization occurred at 170°C. When the temperature reached 180°C, the conversion of PET and the yield of DMT increased dramatically

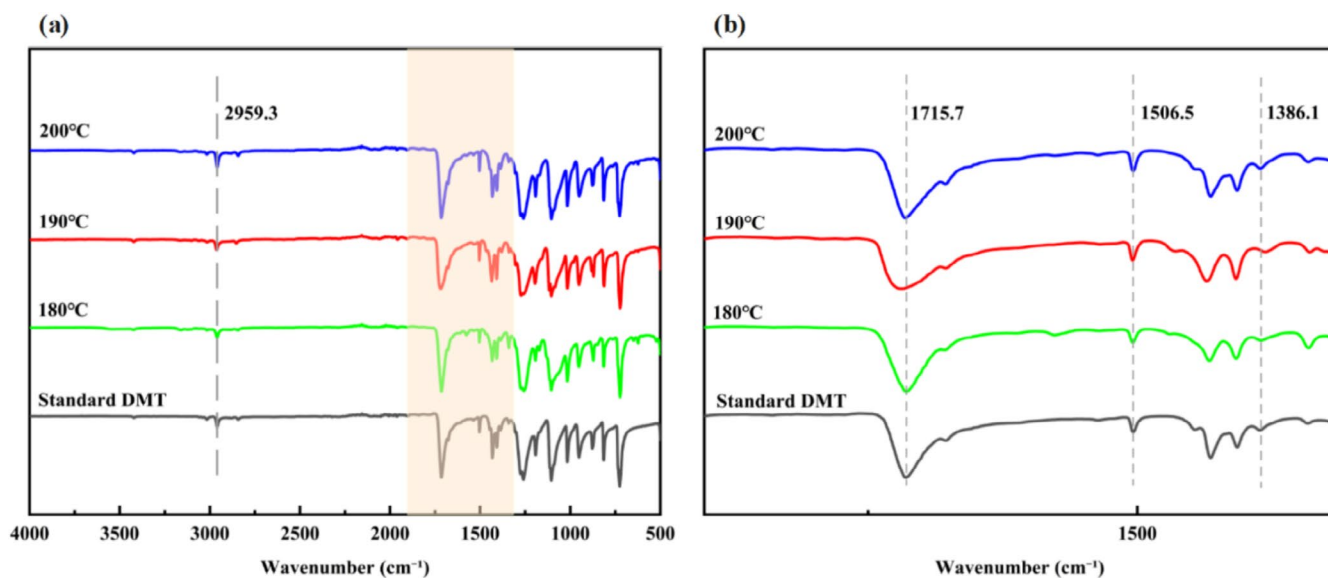


FIGURE 2 | FT-IR spectra of the products (180°C, 190°C, and 200°C) and standard DMT within (a) the global map of wavenumber 500–4000 cm⁻¹ and (b) the orange area. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.70802)]

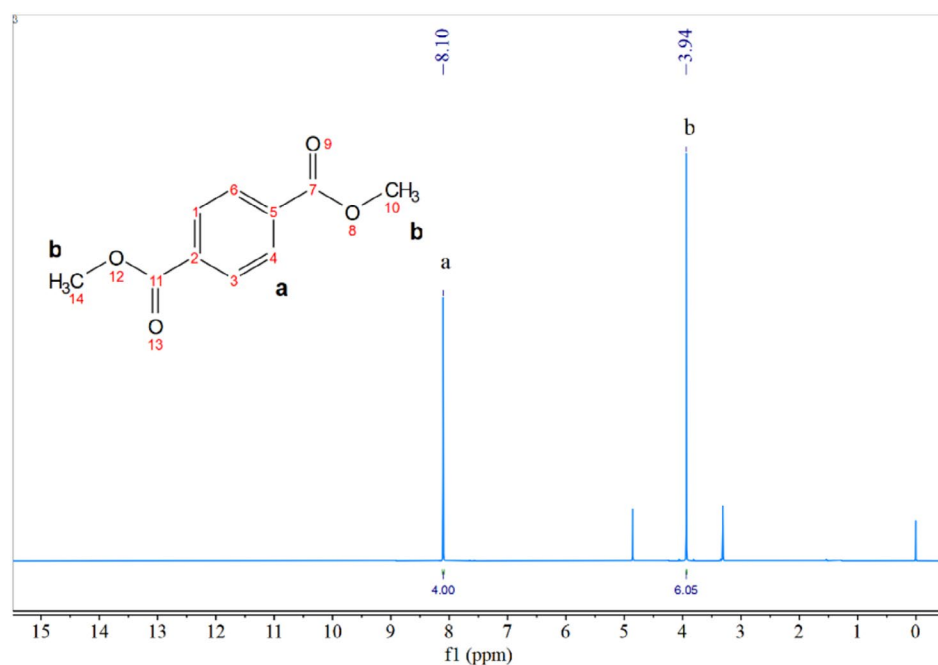


FIGURE 3 | ^1H NMR spectrum of the products (200°C, 2 h, m(PET):m(IL)=5:1, m(PET):m(IL)=1:6). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

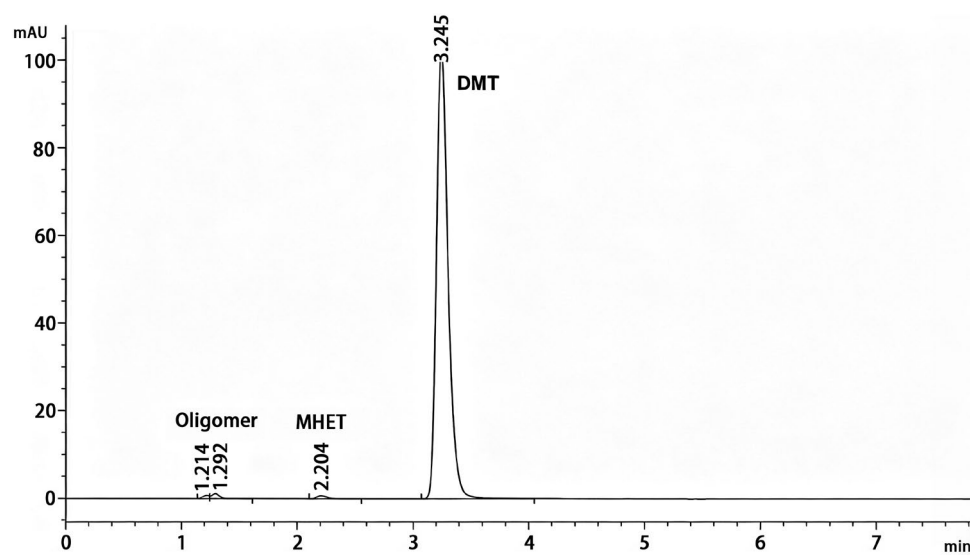


FIGURE 4 | HPLC chromatogram of the products (200°C, 2 h, m(PET):m(IL)=5:1, m(PET):m(IL)=1:6).

to 100% and 55.94%, respectively. With the further increase of temperature, PET was completely depolymerized, and the yield of DMT displayed an initial rapid ascent, followed by slight reduction, reaching a maximum value of 74.06% at 200°C. This trend might be attributed to the enhanced swelling of PET in methanol at elevated temperatures, which facilitates contact between reactants and the catalyst's active sites. However, the yield of DMT decreased to 67.46% when the temperature rose to 220°C, indicating that excessive temperature might promote the side reactions in the reaction system. It could be concluded that temperature played a crucial role in PET methanolysis catalyzed by $[\text{Emim}]\text{BF}_4$. Furthermore, only a minor amount of the intermediate MHET was detected, confirming excellent reaction selectivity in this catalytic system.

As shown in Figure 6b, PET was fully converted after 1 h. The yield of DMT reached a maximum of 74.06% at 2 h and had no further improvement with extended reaction time. In the initial stages, the low concentration of DMT favored the forward reaction. And enough reaction time could enhance PET swelling and improve molecular diffusion over time, which facilitated the degradation kinetics. BHET and MHET content decreased with prolonged reaction time, indicating the gradual conversion to DMT.

The influence of $[\text{Emim}]\text{BF}_4$ dosage on the conversion of PET and the yield of DMT was summarized in Figure 6c. In the absence of a catalyst, the yields of DMT, MHET, and BHET remained below 1%. The addition of $[\text{Emim}]\text{BF}_4$ significantly enhanced the

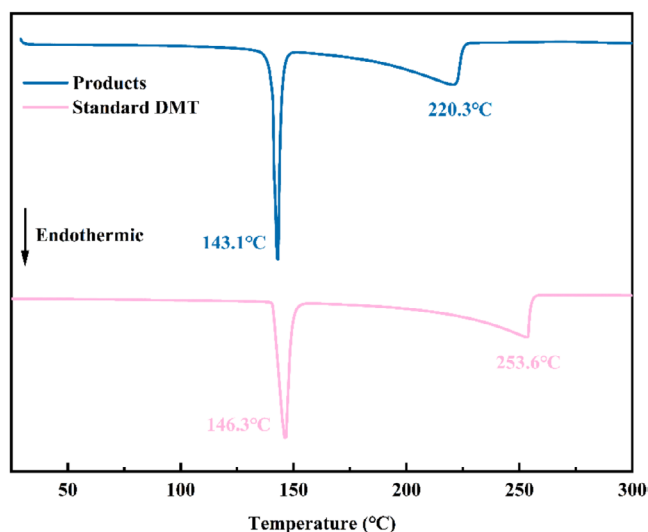


FIGURE 5 | DSC curves of the products and standard DMT (200°C, 2 h, $m(\text{PET}):m(\text{IL})=5:1$, $m(\text{PET}):m(\text{IL})=1:6$). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

reaction. PET completely depolymerized when the mass ratio of PET and $[\text{Emim}]\text{BF}_4$ was 7:1. Further increasing the catalyst loading to a ratio of 5:1 raised the yield of DMT to 74.06%, which could be attributed to the increased number of catalytic active sites. Beyond this point, additional catalyst resulted in a slight decline in the yield of DMT, suggesting that a mass ratio of 5:1 offers sufficient active sites under the given conditions. Throughout these variations, the yield of MHET remained consistently low (5%–6%).

Figure 6d illustrated the influence of methanol dosage on the conversion of PET and the yield of DMT. As the mass ratio of PET to methanol varied from 1:4 to 1:8, PET was completely depolymerized, while the yield of DMT increased first and then decreased, reaching a maximum value of 74.06% when the mass ratio was 1:6. The yield of MHET remained low (5%–6%) throughout. Increasing methanol dosage could enhance PET swelling and reduce the concentration of EG, both favoring the forward reaction. However, beyond the optimal ratio, further increasing methanol leads to catalytic active sites dilution, decreasing the yield of DMT by approximately 10% at a 1:8 ratio. Thus, a mass ratio of PET to methanol of 5:1 was identified as optimal for maximizing the yield of DMT under the investigated conditions.

3.3 | Reaction Kinetics

In the studies of the kinetics of polymer depolymerization, the reaction order is usually considered to be first-order [10–12]. In this study, methanol was always in excess during the reaction. Therefore, it is initially assumed that PET methanolysis in the presence of $[\text{Emim}]\text{BF}_4$ is controlled by the first-order kinetic Equation (3):

$$r = \frac{dC_{\text{PET}}}{dt} = -kC_{\text{PET}} \quad (3)$$

where k represents the rate constant of the reaction and C_{PET} represents the concentration of PET at time t . The concentration of solid PET can be replaced by its mass, while the weight

formation of DMT with reaction time corresponding to that of PET. The reaction rate equation can be rewritten as Equation (4):

$$r = -\frac{d(m_{\text{PET}}/m_{i,\text{PET}})}{dt} = \frac{d(m_{\text{DMT}}/m_{i,\text{DMT}})}{dt} = kf(m_{\text{DMT}}/m_{i,\text{DMT}}) \quad (4)$$

where m_{PET} and m_{DMT} are the mass of PET and DMT at time t , respectively, $m_{i,\text{DMT}}$ is the theoretical mass of DMT produced from the methanolysis of a certain mass of PET, and $m_{i,\text{PET}}$ represents as mentioned above. The yield of DMT can be defined as:

$$Y = \frac{m_{\text{DMT}}}{m_{i,\text{DMT}}} \quad (5)$$

Assuming $f(m_{\text{DMT}}/m_{i,\text{DMT}}) = (m_{\text{DMT}}/m_{i,\text{DMT}})^n$, where n represents the reaction order, then $f(Y) = Y^n$, Equation (4) then can be written as:

$$\frac{dY}{dt} = kY^n \quad (6)$$

When $n = 1$, Equation (6) is integrated over time to give:

$$\ln Y = kt \quad (7)$$

The reaction rate constants at different temperatures obtained by investigating the effects of temperature on the reaction rate of PET methanolysis using $[\text{Emim}]\text{BF}_4$ are shown in Table 1. The reaction rate constants are positively correlated with temperature, indicating that high temperatures favor the reaction.

The activation energy (E_a) can be obtained from the Arrhenius equation:

$$\ln k = -\frac{E_a}{RT} + \ln A \quad (8)$$

where A is the pre-exponential factor, R is the gas constant (8.314 J/(K mol)), and T is the temperature in Kelvin. Figure 7 shows the Arrhenius plot after fitting.

It can be observed that the Arrhenius plot has good linearity. E_a and A of the reaction were calculated to be 51.16 kJ/mol and 2759.21, respectively. The kinetics of the PET methanolysis catalyzed by $[\text{Emim}]\text{BF}_4$ can be effectively described using a reaction kinetic equation:

$$\ln k = -\frac{51.16}{RT} + 7.92 \quad (9)$$

This low activation energy of the $[\text{Emim}]\text{BF}_4$ -based system demonstrates its catalytic advantage in PET methanolysis over other reported approaches summarized in Table 2. While it must be acknowledged that the yield in this study has certain limitations, this IL enables relatively low temperature and short reaction time. Compared with micron-sized particles, the use of waste bottle flakes as substrate demonstrates the high adaptability to diverse raw materials.

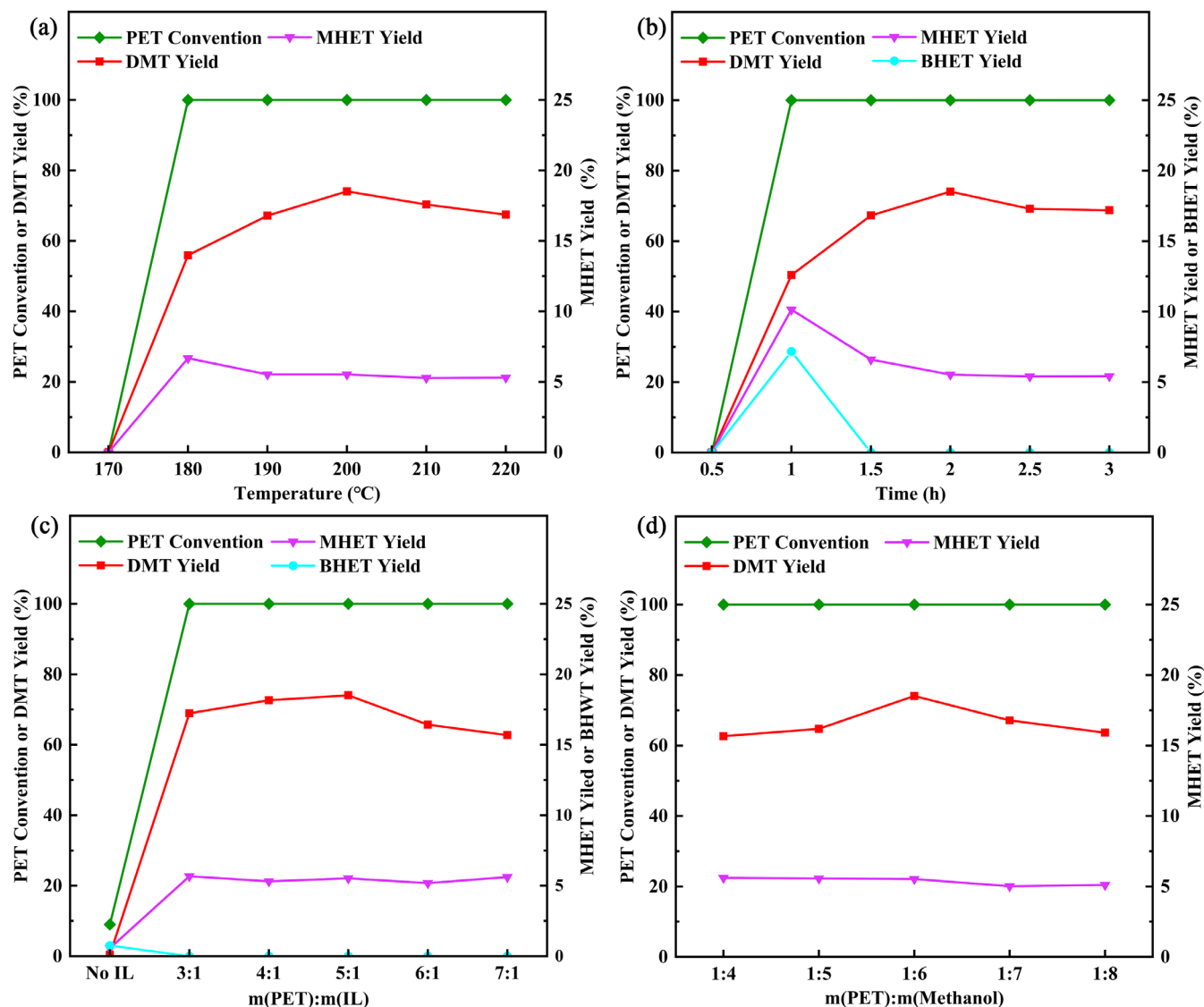


FIGURE 6 | Effects of different reaction parameters on PET methanolysis: (a) temperature, (b) time, (c) catalyst dosage, and (d) methanol dosage. If not specified, the default reaction conditions were as follows: 200°C, 2 h, m(PET):m(IL)=5:1, m(PET):m(Methanol)=1:6. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.70802)]

TABLE 1 | Reaction rate constants at different temperatures.

Temperature (°C)	k (min ⁻¹)	1/T	ln k
180	0.0036	0.002113	-5.0515
190	0.0044	0.002159	-5.4262
200	0.0064	0.002207	-5.6268

3.4 | Proposed Reaction Mechanism and DFT Studies

Having established the optimal conditions for high-yield DMT production, we sought to understand the underlying catalytic mechanism. The morphological evolution of PET during depolymerization, as revealed by SEM (Figure 8), provides the first visual clue at successive stages of depolymerization. The raw PET flakes (Figure 8a) exhibited a relatively smooth and dense surface morphology, consistent with an ordered arrangement of polymer chains. As depolymerization advances (Figure 8b),

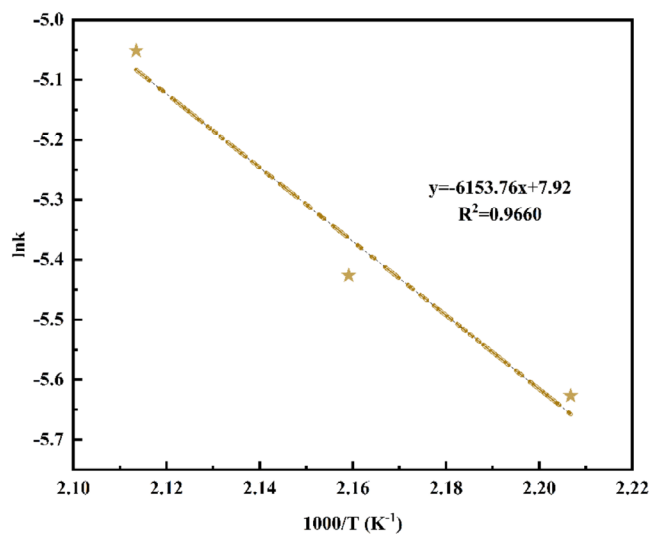
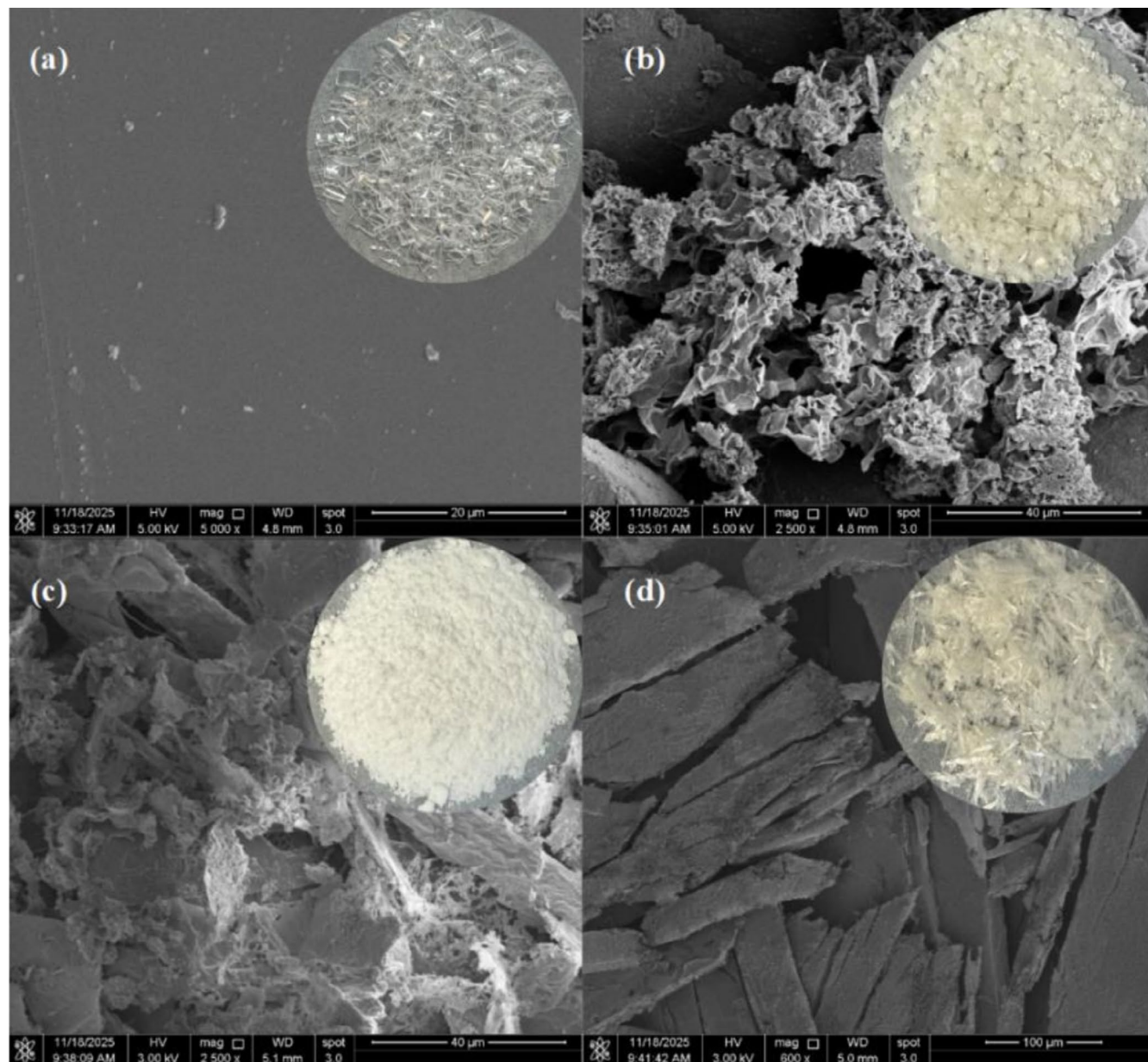
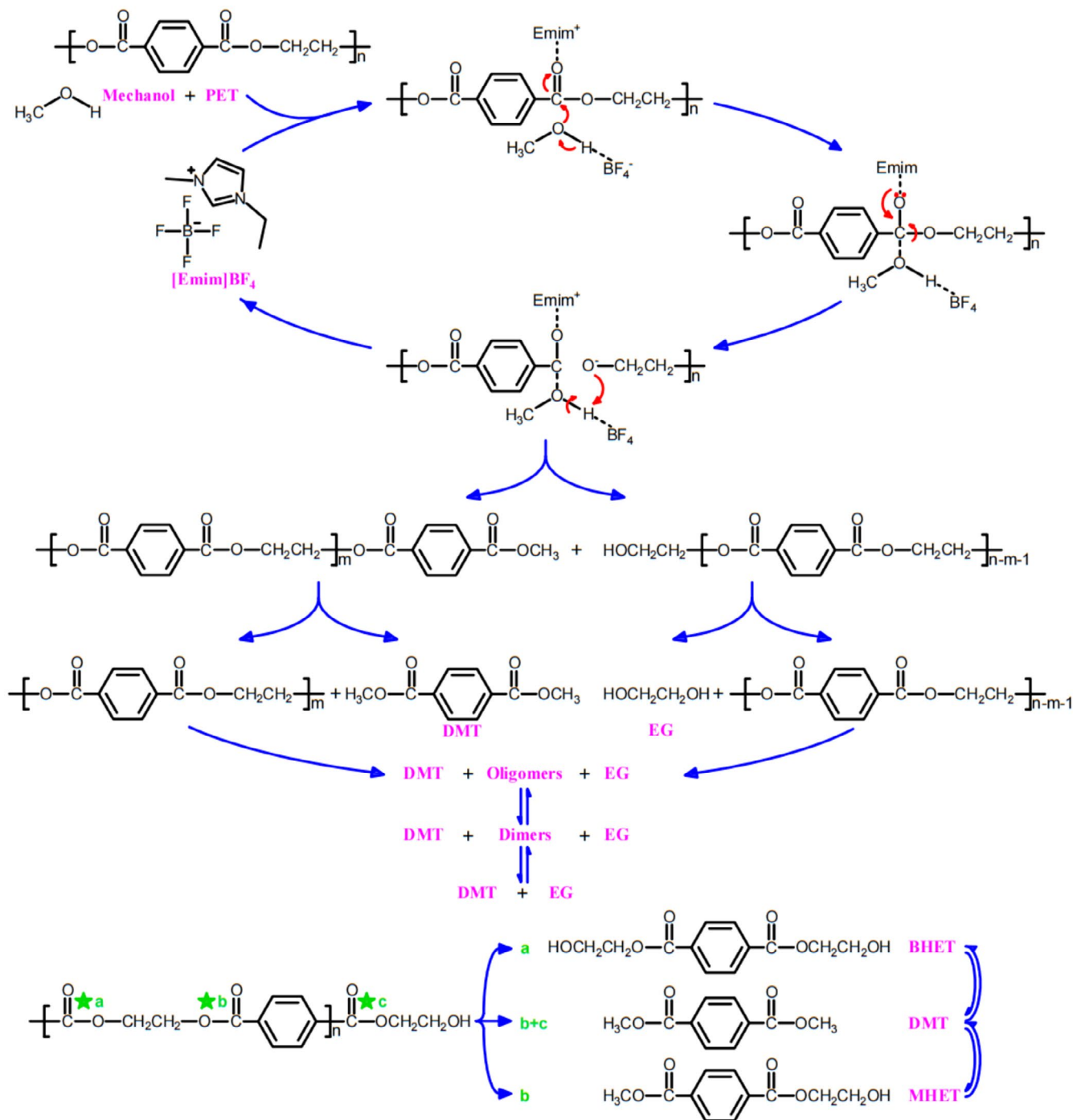


FIGURE 7 | Arrhenius plot of PET methanolysis catalyzed by [Emim]BF₄. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.70802)]

TABLE 2 | Summary of reaction conditions, kinetics, PET conventions and monomer yields of various PET methanolysis techniques.

Catalyst/special conditions	Temperature (°C)	Time (h)	E_a (kJ/mol)	Convention (%)	Yield (%)	References
PILs-Zn ²⁺	170	1	107.6	100	89.1	[7]
K ₂ CO ₃ /cosolvent	25	24	66.5	100	93.1	[8]
CO ₂ -enhanced subcritical CH ₃ OH	220	50 (min)	135.1	100	79.1	[10]
Supercritical CH ₃ OH	310	1	49.9	100	97.7	[26]
Zn(OAc) ₂ /μm-sized PET	120	2	95.31	97.8	97.8	[27]
[Emim]BF ₄	200	2	51.16	100	74.06	This work

**FIGURE 8** | The microstructure of PET raw material and the products under optical microscope. Reaction conditions: (a) raw PET bottle flakes; (b) 180°C, 1 h, m(PET):m(IL)= 5:1, m(PET):m(IL)= 1:6; (c) 200°C, 1 h, m(PET):m(IL)= 5:1, m(PET):m(IL)= 1:6; (d) 200°C, 2 h, m(PET):m(IL)= 5:1, m(PET):m(IL)= 1:6. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]



SCHEME 1 | Proposed mechanism for PET methanolysis catalyzed by [Emim]BF₄ [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.70802)]

methanol and [Emim]BF₄ penetrated between the PET molecules, disrupting the initial structural order. Swelling of PET further facilitated diffusion and weakened intermolecular interactions [28]. Subsequently, under the combined action of the ionic liquid and methanol, PET depolymerized into shorter chains. In later stages (Figure 8c,d), the monomer DMT gradually formed in the products, exhibiting a disordered, irregular, and rough-edged block-like morphology.

Based on these morphological changes and the known reactivity of ionic liquids, we propose a synergistic catalytic mechanism

(Scheme 1). The [Emim]⁺ cation activates the carbonyl group of PET, protonating the ester carbon atom of PET and enhancing its electrophilicity. The anion BF₄⁻ forms hydrogen bonds with the hydroxyl hydrogen of the methanol, which enhances the electronegativity of the hydroxyl oxygen and facilitates its nucleophilic attack on the PET ester carbon cation [29, 30]. With the synergy of these two effects, the nucleophile attacks the ester group of PET, leading to the cleavage of the C–O bond [24]. Subsequently, a new C=O bond is formed after the hydroxyl –OH of methanol breaks and electron transfers from –O[Emim]⁺ to the ester carbon. A new C–O bond is established between the methanol

hydroxyl oxygen and the acyl carbon of PET, while the hydrogen released from methanol combines with the $-OCH_2CH_2-$ group at the other cleaved end of PET to form $HOCH_2CH_2-$ [14]. After the ionic liquid is regenerated, $[Emim]^+$ and BF_4^- are released upon break of the hydrogen bonds and recombine into the original structure, ready to re-enter the catalytic cycle. This

depolymerization proceeds progressively along the PET chain, gradually breaking it down from long polymers into oligomers, dimers, and finally monomers. An equivalent molar amount of EG is released along with DMT. Since depolymerization can occur randomly at the ester groups on both sides of the benzene ring, the production of DMT occurs selectively, accompanied by the formation of minor amounts of intermediates (MHET and BHET). Moreover, a reversible reaction between EG and DMT may form a small amount of intermediates, eventually reaching equilibrium [7].

To validate the proposed mechanism and quantify the catalytic effect of $[Emim]BF_4$, DFT calculations were performed. The overall reaction process was hypothesized to proceed in two stages, wherein methanol and the ionic liquid sequentially attacked the ester groups located at the outer and inner sides of the terminal benzene ring in the PET chain, ultimately yielding the target product DMT. Through geometry optimization and transition-state searches, the key configurations involved in the PET methanolysis process in the presence of $[Emim]BF_4$ were obtained (Figures 9 and 10). In the initial configuration (IS1 and IS2), hydrogen bonds were formed between the BF_4^- anion and the methanol molecule. This interaction was suggested to increase the electronegativity of the oxygen atom by modulating the local charge density, enhancing the nucleophilic ability of the hydroxyl oxygen toward the ester group. The transition states (TS1 and TS2) corresponded to the cleavage of the C–O bond in the PET ester group induced by nucleophilic attack from the methanol hydroxyl oxygen. Notably, an intermediate state (IMS) was observed in the second reaction stage, which was optimized to ultimately form the final state (FS2). $[Emim]BF_4$ was

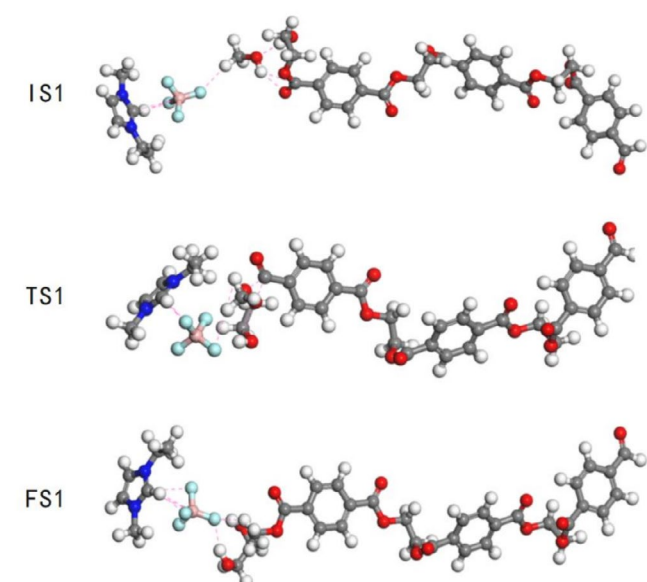


FIGURE 9 | The configurations of the PET methanolysis process in the first stage with $[Emim]BF_4$. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

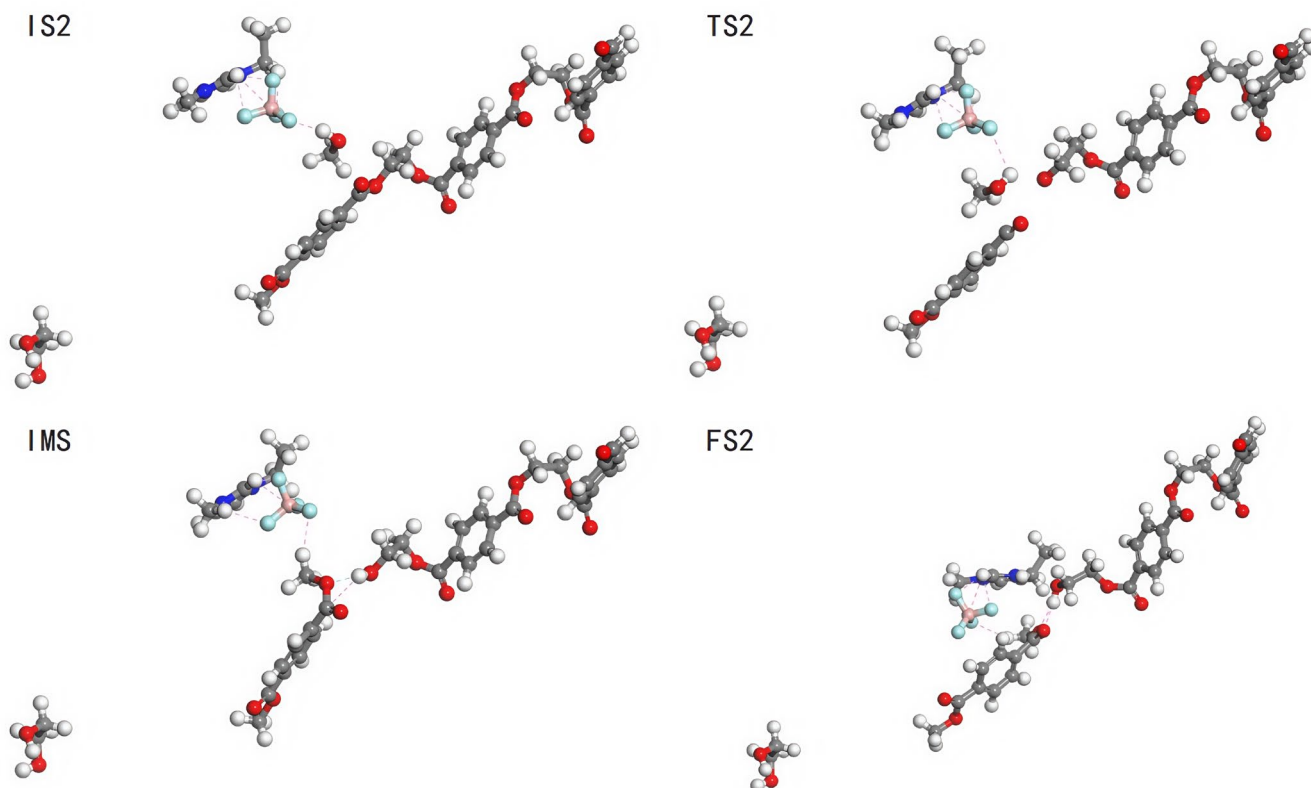


FIGURE 10 | The configurations of the PET methanolysis process in the second stage with $[Emim]BF_4$. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

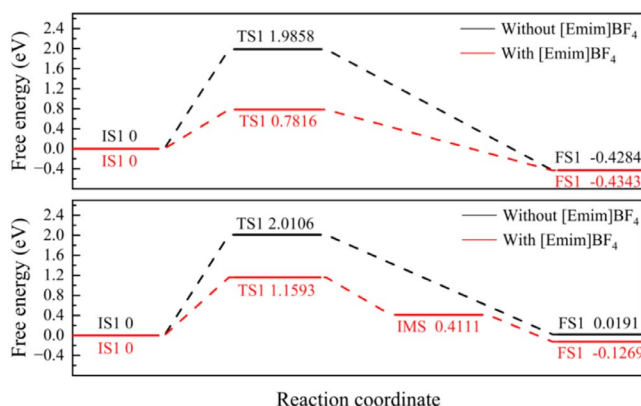


FIGURE 11 | The energy change of PET methanolysis reaction with or without [Emim]BF₄. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.70802)]

reintegrated to with its structural integrity maintained after the completion of the reaction.

The energy profiles of PET methanolysis with and without the involvement of [Emim]BF₄ are illustrated in Figure 11, and the detailed energy data in different states are summarized in Tables S1 and S2. The energy barriers corresponding to TS1 and TS2 were markedly reduced from 1.9858 and 2.0106 eV to 0.7816 and 1.1593 eV, respectively. TS2 with higher overall energy was considered the rate-determining step of the reaction. The observed lowering of the energy barriers corresponding to TS2 implied a reduced energetic obstacle for the process, which in turn facilitated an increase in the reaction rate. Although IMS in the second stage lay energetically higher than the initial state IS2, the overall energy change between IS2 and FS2 shifted from an endergonic value of +0.0191 eV to an exergonic value of −0.1269 eV with the inclusion of [Emim]BF₄. This change indicated that the reaction product became more stabilized and the process was thermodynamically more favorable.

4 | Conclusion

In summary, this study demonstrated that [Emim]BF₄ served as an efficient catalyst for the methanolysis of PET. Comprehensive characterization confirmed that the main component of the depolymerization products is DMT, and [Emim]BF₄ exhibited good thermal stability throughout the process. Complete depolymerization of PET was achieved, with a DMT yield of 74.06% under the optimal reaction conditions (a reaction temperature of 200°C, a reaction time of 2 h, a PET-to-[Emim]BF₄ mass ratio of 5:1, and a PET-to-methanol mass ratio of 1:6). Kinetic analysis revealed that the reaction follows first-order kinetics, with an apparent activation energy of 51.16 kJ/mol. Furthermore, a mechanistic insight into the methanolysis of PET was elucidated. According to DFT calculations, [Emim]BF₄ can accelerate the reaction rate and favor the forward process by reducing the energy barriers and stabilizing the final state. Overall, this efficient depolymerization process offers a promising and sustainable solution for chemical recycling of waste PET, contributing to a circular plastics economy.

Author Contributions

Wei Du: writing – review and editing, visualization, validation, writing – original draft, conceptualization. **Jing Xie:** validation, methodology, investigation. **Jianlei Qi:** software, methodology. **Dan Zhou:** writing – review and editing, supervision, data curation. **Jianzhong Yin:** supervision, project administration, funding acquisition, formal analysis.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Experimental apparatus used for degradation of PET. (1) Thermocouple; (2) Reactor; (3) Pressure gauge; (4) Oil bath; (5) Controller; (6) Needle valve. **Figure S2:** ¹H NMR spectra of DMT standard sample. **Figure S3:** HPLC chromatogram of the standard DMT. **Figure S4:** HPLC chromatogram of the standard MHET. **Figure S5:** HPLC chromatogram of the standard BHET. **Figure S6:** PET configuration. **Figure S7:** The configurations of the PET methanolysis process without [Emim]BF₄. **Table S1:** The energy and its changes in different states without [Emim]BF₄. **Table S2:** The energy and its changes in different states with [Emim]BF₄.